1/9/26

Practical-5

Aim: Experiments on Packet capture tool: Wireshark.

Packet Sniffer:

Sniffs messages being sent/received from/by your computer.

Store & display the contents of the various protocol fields in the messages

Passive Program
- never sends packets itself
- no packets addressed to it
- receives a copy of all packets (sent/received)

Packet Sniffer structure Diagnostic Tool:

- Tcpdump
  - tcpdump enx host 10.129.41.2 -w exe3.out
- Wireshark
  - wireshark -r exe3.out

Description:

Wireshark - a network analysis tool formerly known as Ethereal, captures packets in real time & displays them in human-readable format.

It is used for:

- Network administrators: troubleshoot network problems
- Network security engineers: examine security problems
- Developers: debug protocol implementations
- People: learn network protocol internals.

Getting Wireshark:

Wireshark can be downloaded for windows or macOS from its official website. For linux or another UNIX-like system, Wireshark will be found in its package repositories.

Capturing Packets

After downloading & installing Wireshark, launch it & double-click the name of a network interface under Capture to start capturing packets on that interface.

Packet List Pane:

It displays all the packets in the current capture file. In "Packet list" pane each line in the packet list corresponds to one packet in the capture file.

Packet Details Pane:

It shows the current packet in a more detailed form. This pane shows the protocols & protocol fields of packet selected in the "Packet List" pane.

Packet Bytes: It shows the data of the current packet in a hexdump style.

Color Coding - Wireshark uses colors to identify types of traffic at a glance.

Sampling Captures - Wiki contains a page of sample capture files that you can load & inspect.

Capturing & Analysing packets using Wireshark tool

To filter, capture, view, packets in Wireshark tool.
Capture 100 packets from The Ethernet/IEEE 802.3 LAN interface & save it.

Procedure:

Select Local Area Connection in Wireshark

Go to capture option

Select stop capture automatically after 100 packets

Then click Start capture

Save the packets

1. Create a Filter to display only TCP/UDP packets, Inspect the packets & provide the flow graph

Procedure:

→ Select Local Area Connection in Wireshark.

→ Go to capture option

→ Select stop capture automatically after 100 packets.

→ Then click start capture.

→ Search TCP packets in search bar.

→ To see flow graph click statistics flow graph.

→ Save the packets.

2. Create a filter to display only ARP packets & inspect the packets.

Procedure:

Go to capture option

Select stop capture automatically after 100 packets.

Then click start capture

Search ARP packets in search bar

Save the packets

3. Create a filter to display only DNS packets & provide the flow graph.

→ Go to capture option
→ Select stop automatically after 100 packets
→ Then click start capture
→ Search DNS packets in search box
→ To see flow graph click statistics flow graph.
→ Save the packets

4. Create a filter to display only HTTP packets & inspect the packets

Procedure:

→ Select Local Area Connection in Wireshark.

→ Go to capture option
→ Select stop capture automatically after 100 packet.
→ Then click start capture.
→ Search HTTP packets in Search bar.
→ Save the packets.

Create a filter to display only IP/ ICMP packets & inspect the packets.

Procedure:

→ Select local area connection in Wireshark.
→ Go to capture option
→ Select stop capture automatically after 100 packet
→ Then click start capture.
→ Search ICMP/IP packets in Search bar.

6. Create a filter to display only DHCP packets & Inspect the packets.

Procedure:

Select local area connection in wireshark.

Go to capture option

Select stop a capture automatically after 100 packets.

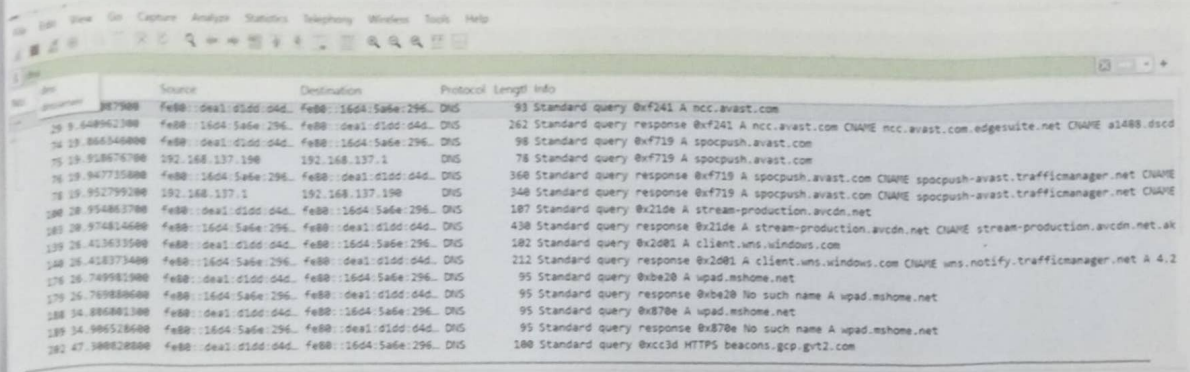
Then click start capture.

Search DHCP packets in search bar.

Save the packets.

Result: Thus, packets were captured using wireshark tool.

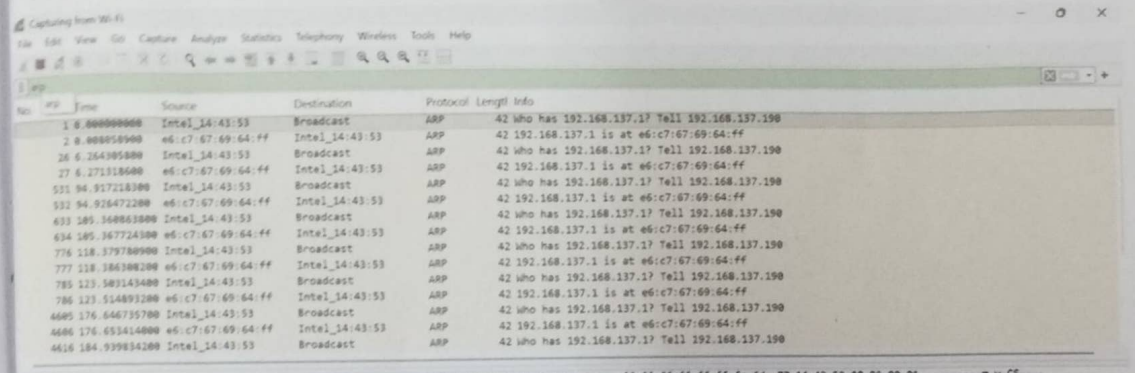
. r. to capture option



No.	Source	Destination	Protocol	Length	Info
29	fe80::dea1:d1dd:d4d...	fe80::16d4:5a6e:296...	DNS	93	Standard query 0xf241 A ncc.avast.com
74	fe80::16d4:5a6e:296...	fe80::dea1:d1dd:d4d...	DNS	262	Standard query response 0xf241 A ncc.avast.com CNAME ncc.avast.com.edgesuite.net CNAME a1488.dscd
75	fe80::dea1:d1dd:d4d...	fe80::16d4:5a6e:296...	DNS	98	Standard query 0xf719 A spocpush.avast.com
76	192.168.137.190	192.168.137.1	DNS	78	Standard query 0xf719 A spocpush.avast.com
78	fe80::16d4:5a6e:296...	fe80::dea1:d1dd:d4d...	DNS	360	Standard query response 0xf719 A spocpush.avast.com CNAME spocpush-avast.trafficmanager.net CNAME
79	192.168.137.1	192.168.137.190	DNS	348	Standard query response 0xf719 A spocpush.avast.com CNAME spocpush-avast.trafficmanager.net CNAME
100	fe80::dea1:d1dd:d4d...	fe80::16d4:5a6e:296...	DNS	107	Standard query 0x21de A stream-production.avcdn.net
101	fe80::16d4:5a6e:296...	fe80::dea1:d1dd:d4d...	DNS	430	Standard query response 0x21de A stream-production.avcdn.net CNAME stream-production.avcdn.net.ak
139	fe80::dea1:d1dd:d4d...	fe80::16d4:5a6e:296...	DNS	102	Standard query 0x2d01 A client.wns.windows.com
140	fe80::16d4:5a6e:296...	fe80::dea1:d1dd:d4d...	DNS	212	Standard query response 0x2d01 A client.wns.windows.com CNAME wns.notify.trafficmanager.net A 4.2
176	fe80::dea1:d1dd:d4d...	fe80::16d4:5a6e:296...	DNS	95	Standard query 0xbe20 A wpad.mshome.net
179	fe80::16d4:5a6e:296...	fe80::dea1:d1dd:d4d...	DNS	95	Standard query response 0xbe20 No such name A wpad.mshome.net
188	fe80::dea1:d1dd:d4d...	fe80::16d4:5a6e:296...	DNS	95	Standard query 0x870e A wpad.mshome.net
189	fe80::16d4:5a6e:296...	fe80::dea1:d1dd:d4d...	DNS	95	Standard query response 0x870e No such name A wpad.mshome.net
282	fe80::dea1:d1dd:d4d...	fe80::16d4:5a6e:296...	DNS	180	Standard query 0xcc3d HTTPS beacons.gcp.gvt2.com

```
Frame 28: Packet, 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interfa
Ethernet II, Src: Intel_14:43:53 (fc:6d:77:14:43:53), Dst: e6:c7:67:69:64:ff (e6:c7:67
Internet Protocol Version 6, Src: fe80::dea1:d1dd:d400:7d2a, Dst: fe80::16d4:5a6e:296b
User Datagram Protocol, Src Port: 63236, Dst Port: 53
Domain Name System (query)
```

```
0000  e6 c7 67 69 64 ff fc 6d  77 14 43 53 86 dd 60 00   gid  m w CS  `
0010  5d 99 00 27 11 80 fe 80  00 00 00 00 00 00 de a1   ]  '
0020  d1 dd d4 d0 7d 2a fe 80  00 00 00 00 00 00 16 d4    }*
0030  5a 6e 29 6b ab ef f7 04  00 35 00 27 46 52 f2 41   Zn)k       5 'FR A
0040  01 00 00 01 00 00 00 00  00 00 03 6e 63 63 05 61   ncc a
0050  76 61 73 74 03 63 6f 6d  00 00 01 00 01           vast com
```



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	Intel_14:43:53	Broadcast	ARP	42	Who has 192.168.137.1? Tell 192.168.137.190
2	0.000058900	e6:c7:67:69:64:ff	Intel_14:43:53	ARP	42	192.168.137.1 is at e6:c7:67:69:64:ff
26	6.264305800	Intel_14:43:53	Broadcast	ARP	42	Who has 192.168.137.1? Tell 192.168.137.190
27	6.271318600	e6:c7:67:69:64:ff	Intel_14:43:53	ARP	42	192.168.137.1 is at e6:c7:67:69:64:ff
531	94.917218300	Intel_14:43:53	Broadcast	ARP	42	Who has 192.168.137.1? Tell 192.168.137.190
532	94.926472200	e6:c7:67:69:64:ff	Intel_14:43:53	ARP	42	192.168.137.1 is at e6:c7:67:69:64:ff
633	105.360863800	Intel_14:43:53	Broadcast	ARP	42	Who has 192.168.137.1? Tell 192.168.137.190
634	105.367724300	e6:c7:67:69:64:ff	Intel_14:43:53	ARP	42	192.168.137.1 is at e6:c7:67:69:64:ff
776	118.379780900	Intel_14:43:53	Broadcast	ARP	42	Who has 192.168.137.1? Tell 192.168.137.190
777	118.386308200	e6:c7:67:69:64:ff	Intel_14:43:53	ARP	42	192.168.137.1 is at e6:c7:67:69:64:ff
785	123.503143400	Intel_14:43:53	Broadcast	ARP	42	Who has 192.168.137.1? Tell 192.168.137.190
786	123.514893200	e6:c7:67:69:64:ff	Intel_14:43:53	ARP	42	192.168.137.1 is at e6:c7:67:69:64:ff
4605	176.646735700	Intel_14:43:53	Broadcast	ARP	42	Who has 192.168.137.1? Tell 192.168.137.190
4606	176.653414000	e6:c7:67:69:64:ff	Intel_14:43:53	ARP	42	192.168.137.1 is at e6:c7:67:69:64:ff
4616	184.939834200	Intel_14:43:53	Broadcast	ARP	42	Who has 192.168.137.1? Tell 192.168.137.190

```
Frame 1: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interfac
Ethernet II, Src: Intel_14:43:53 (fc:6d:77:14:43:53), Dst: Broadcast (ff:ff:ff:ff:ff:f
Address Resolution Protocol (request)
```

```
0000  ff ff ff ff ff ff fc 6d  77 14 43 53 08 06 00 01   m w CS
0010  08 00 06 04 00 01 fc 6d  77 14 43 53 c0 a8 89 be   m w CS
0020  00 00 00 00 00 00 c0 a8  89 01
```

